

allows the production of larger silicon ingots, which are essential for microelectronic IC fabrication. However, since the material comes into contact with a quartz crucible, any contamination in the crucible can affect the ingot's quality. To mitigate this, strict quality control is applied to both the crucible and raw materials. Despite these efforts, some trade-offs are inevitable between achieving high purity and producing larger-diameter wafers to meet the semiconductor industry's demands.

How does the Czochralski technique help reduce impurities in crystal manufacturing?

To reduce impurities during Czochralski processing, a magnetic field is applied to lower oxygen concentrations in the wafer. The field drives impurities to the crucible's base. Using this method, we produce specialised laser crystals intended for supply to the Ministry of Defence's Ordnance Factory.

How does the pulling rate impact crystal quality, and why is it critical for wafers?

Crystal pulling speed directly affects quality. Silicon grows at about 1mm per minute, whereas oxide crystals like YAG require a much slower rate—around 0.5mm per hour. Exceeding the optimal pulling speed for YAG crystal, such as pulling faster than 0.5mm per hour, can disrupt the formation of a single crystal. This can lead to the inclusion of polycrystalline structures, which degrade the crystal's quality. Silicon ingots with dislocations are likely to get rejected. Therefore, maintaining the quality of a single crystal throughout the entire 3-metre ingot is essential. Without proper crystal quality, many wafers may be rejected after slicing and wafering due to polycrystalline inclusions.

Why is the Czochralski technique preferred for large-diameter silicon wafer production?

The Czochralski technique is the only method capable of producing large-diameter silicon ingots with the desired crystal orientation, matching the ideal single-crystal orientation of the seed. In this process, the oriented seed is dipped into the molten silicon to initiate crystal pulling, ensuring consistent orientation throughout the crystal. Alternative methods that do not use a seed result in random crystal orientations. Identifying the desired orientation and extracting the correct diameter from these crystals becomes complex and tedious. This is why the Czochralski technique is preferred for silicon wafers, especially for diameters larger than 203mm. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/efyNov2025>



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